Software Requirements Specification

for

SkipQ

Version 1.1

Prepared by

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Course: CS253

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Aayush Gajeshwar Jashan Mittal Harpuneet Singh Sourabh Sankhala Surya Prakash Pingali Aarya Bhatt Pratham Sachin Todkar Jarapa Ashish Pamar Jatin Toshiwal Usnish Paul	First Version of the Software Requirement Specifications Document of SkipQ	22/01/2026
1.1	Aayush Gajeshwar Jashan Mittal Harpuneet Singh Sourabh Sankhala Surya Prakash Pingali Aarya Bhatt Pratham Sachin Todkar Jarapa Ashish Pamar Jatin Toshiwal Usnish Paul	Worked on changes suggested by mentor.	24/01/2026

1. Introduction

1.1 Product Scope

SkipQ is a canteen automation and management service for IIT Kanpur campus. It offers a single centralized platform for everyone on campus to order or pre-order food from any canteen on campus as well as save time by offering information about waiting times in each canteen. It also offers canteen managers an interface to manage the operations of the canteen easily and effectively.
The software has three interfaces: for canteen managers, customers, and admins.
On the canteen managers' side, the software can be used to see a list of current pending orders as well as set the expected waiting time for other customers to see when they order using the software.
On the admin's side, the software allows system oversight, canteen onboarding, and user management to maintain platform integrity.
On the customer's side, the software can be used to pre-book the canteen products as well as the expected waiting time that each order can take in each canteen. The user can also see the previous ratings of each product which have been given by other customers in the past to help them judge the quality of a new product before trying it out.
SkipQ is a powerful time saving tool for the customers in the campus as it not only provides a convenient centralized platform to order products from any canteen but also offers vital information about each product such as ratings and expected waiting time which makes the canteen experience much better.

1.2 Intended Audience and Document Overview

This document serves as the primary reference for all project stakeholders, including developers, project managers, and quality assurance teams. It comprehensively defines the functional specifications, non-functional requirements, and system constraints necessary to execute a successful development and deployment strategy.

Document Overview

Introduction: Outlines document conventions and abbreviations. While experienced readers may bypass this section, it serves as a vital glossary for clarifying terminology.

Overview of the system: Summarizes the software's functionality, assumptions, and dependencies. It provides a high-level context, establishing a foundation for the detailed technical sections.

Specific requirements: Details specific features and functions, often visualized with tree diagrams. This section acts as a primary blueprint for developers and a reference manual for end-users.

Non-Functional requirements: Defines critical system attributes such as performance, reliability, and scalability. This guidance is essential for developers to ensure technical quality.

Other Requirements: Covers essential constraints including security, compliance, compatibility, and integration. These factors are critical for overall success but fall outside standard functional categories.
Appendix: Contains supplementary resources, such as the Data Dictionary for database elements and the Group Log for tracking decisions. It ensures consistency and clarity for all stakeholders throughout the project lifecycle.

1.3 Definitions, Acronyms and Abbreviations

The definitions, acronyms and abbreviations used in this document are as follows -

- **API:** Application Programming Interface – Set of protocols for communication between different parts of the SkipQ system or with external systems.
- **Dashboard:** A personalized interface showing order history, ongoing bookings, and account details.
- **DBMS:** Database Management System. Software used to store, retrieve, and manage data in databases for the SkipQ system.
- **Event Booking:** A feature in SkipQ that allows users to reserve space and arrange catering for special events on campus.
- **HTTPS:** Hypertext Transfer Protocol Secure
- **Live Updates:** Real-time information provided to users about their orders, including preparation, estimated delivery time, and completion status.
- **ML:** Machine Learning
- **OS:** Operating System
- **Peak Periods:** High-demand times during which the canteen experiences a surge in orders.
- **PIN:** Personal Identification Number. A secret numeric code (typically 4 to 6 digits) used as an additional security layer to verify a user's identity for financial transactions, accessing devices, or online accounts.
- **Pre-order Meals:** The process of ordering meals in advance to ensure availability and reduce wait times.
- **UI:** User Interface. The part of the system with which the user interacts.
- **Admin:** System Administrator. A super-user with high-level access to manage canteens, users, and global settings.
- **UPI:** Unified Payments Interface, used for cashless transactions.

- **WebApp: Web Application –** A browser-based application providing the SkipQ services to users, accessible via internet-enabled devices.

1.4 Document Conventions

- This document follows the IEEE standards for Software Requirements Specification (SRS).
- The document is written with the Arial font of size 11 with single spacing and 1 inch margins.
- Important keywords are highlighted in bold font and comments are italicized.
- Section titles are hierarchical and numbered, with clear and concise descriptions under headings.

1.5 References and Acknowledgments

We would like to thank the following people for their continuous support and guidance throughout the making of this document -

- Professor Indranil Saha
- Our TA, Mr Shaikh Imran

2. Overall Description

2.1 Product Overview

SkipQ is a new and self-contained canteen automation and management system developed for the IIT Kanpur campus. It is not a follow-on product of any existing software system, nor is it meant to directly replace a single legacy system. Instead, SkipQ aims to improve and digitize the mostly manual and decentralised food ordering and queue management processes that are currently followed by campus canteens. These existing processes often lead to long waiting times, unclear order queues, and inefficient coordination between customers and canteen staff. SkipQ addresses these issues by providing a centralized digital platform for both customers and canteen managers.

From a system point of view, SkipQ works as a campus-wide service with three main user interfaces: one for customers, one for canteen managers and one for admins. The customer interface allows users to browse menus from different canteens, pre-order food items, view the expected waiting time for each order, and check ratings given by previous customers. This helps users make better decisions before placing an order. On the other hand, the canteen manager interface allows canteen staff to view current and pending orders, manage order queues, and update estimated waiting times, which are then displayed to customers in real time.

SkipQ can also be seen as a part of the larger campus digital ecosystem and may later be integrated with services such as user authentication, online payment systems, and notification services. Overall, SkipQ acts as a bridge between customers and canteens, helping reduce physical crowding, save time, and improve the overall canteen experience on campus.

2.2 Product Functionality

SkipQ provides a centralized platform that supports both customers and canteen managers in efficiently managing food ordering and queue operations across the IIT Kanpur campus. At a high level, the major functions of the system include the following:

- Allow customers to browse menus from multiple campus canteens through a single platform.
- Enable customers to place food orders and pre-orders from selected canteens.
- Display estimated waiting times for orders from each canteen before and after order placement.
- Allow customers to view ratings and feedback for different food items based on previous user experiences.
- Maintain a record of customer orders for reference and order tracking.
- Provide canteen managers with access to a dashboard showing current, pending, and completed orders.
- Allow canteen managers to manage and update order queues in real time.

- Enable canteen managers to set and modify expected waiting times for customer orders.
- Store and manage data related to orders, waiting times, and product ratings in a centralised system.
- Facilitate communication between customers and canteen managers through system updates and status changes.
- Admins can verify and onboard new canteens to the platform.
- Enable Admins to monitor global platform activity, revenue, and system health.

2.3 Design and Implementation Constraints

The development process adheres to the following technical and structural limitations:

- **Hardware and OS:** Backend requires a UNIX-based operating system. Canteen manager-side must function on standard consumer devices without demanding high processing power. Server memory must support ML batch processing alongside live transactions.
- **Database Technologies:** Storage is restricted to PostgreSQL or MongoDB. The schema must accommodate large historical datasets necessary for training future machine learning models.
- **Payment System Staging:** Deployment follows a two-phase strategy: initially, a simulated fake wallet, followed by Razorpay API/Other UPI app integration. The code structure must be modular to support this transition.
- **Machine Learning Architecture:** The system design must support future inference models, including pipelines for feeding order history into recommendation engines and wait-time predictors.
- **Network Protocols:** Continuous internet connectivity is required; offline synchronisation is not supported. All communications must use HTTPS, with WebSockets or efficient polling for real-time queue updates.
- **Security Standards:** Student registration is restricted to valid IIT Kanpur email domains. Manager registration allows standard email providers but requires verification. Passwords must use standard cryptographic encryption. Financial data handling must eventually comply with standard payment security protocols.
- **Concurrent Operations:** The architecture must support a minimum of 500 simultaneous users, dictating specific load-balancing strategies and server resource allocation.
- **Browser Compatibility:** The frontend targets modern web browsers running on smartphones and laptops (Chrome, Firefox, Safari, Edge). Support for legacy browsers is excluded.

2.4 Assumptions and Dependencies

The project relies on the following assumptions and external factors:

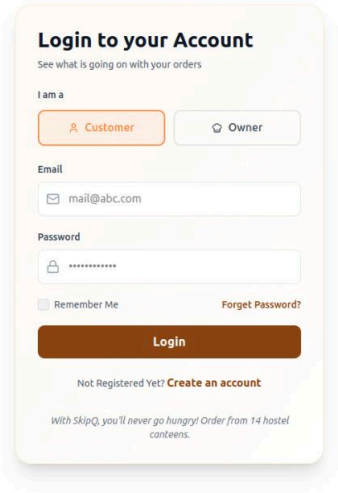
- **Network Stability:** The system assumes the campus network provides consistent, high-speed connectivity to both hostels and canteen locations.
- **Third-Party Payment APIs:** The project assumes Razorpay's/UPIs sandbox and live environments will remain stable and consistent throughout development.
- **Institutional Infrastructure:** User verification depends entirely on the uptime of the IIT Kanpur email server.
- **ML Data Volume:** Accurate recommendations and wait-time predictions assume a sufficient volume of historical order data will be generated.
- **Library Maintenance:** The project assumes selected open-source libraries (UI, Database, ML) will receive security updates and remain compatible with the core framework.
- **Device Availability:** The design assumes the user base possesses smartphones or laptops capable of running modern, JavaScript-enabled web applications.

3. Specific Requirements

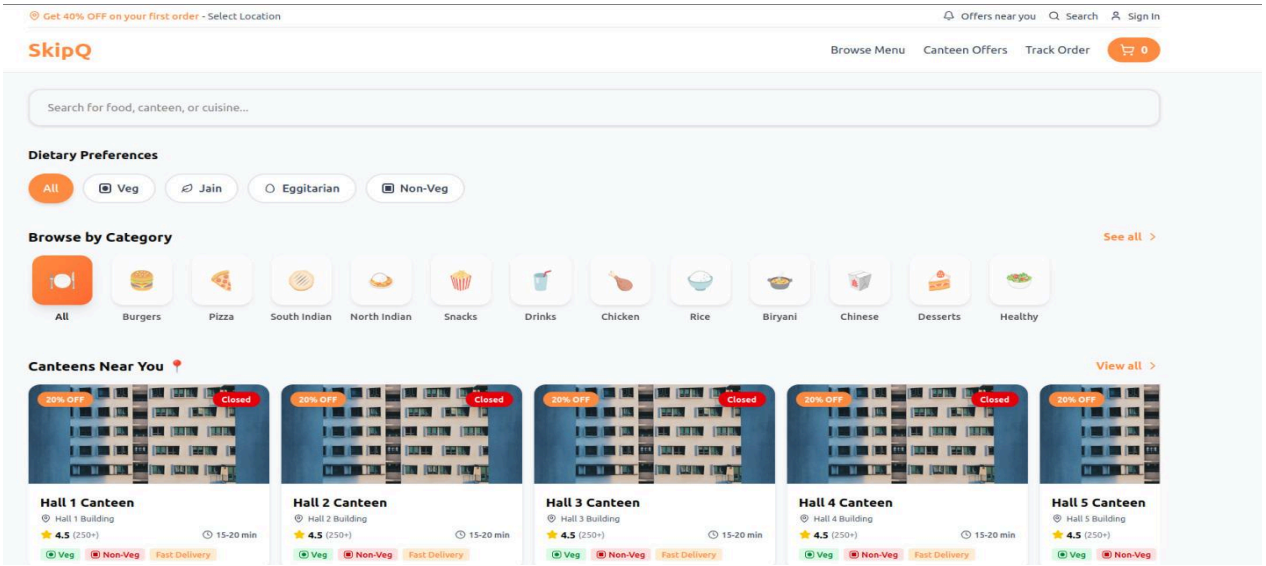
3.1 External Interface Requirements

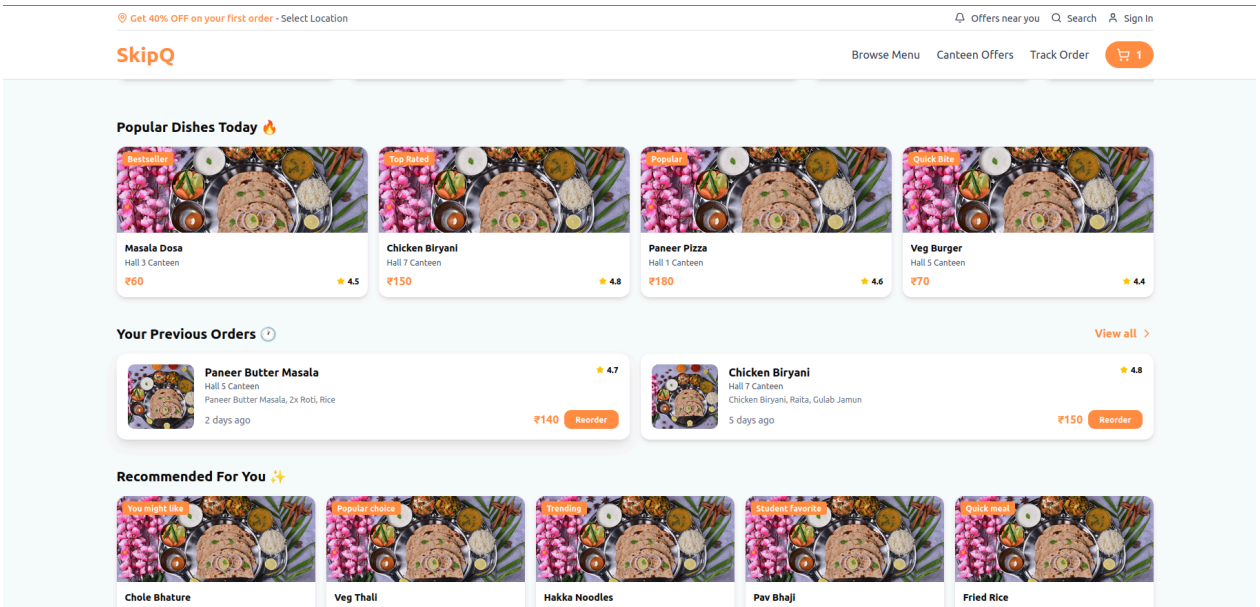
3.1.1 User Interfaces

- The application offers login/signup interfaces for both the canteen managers/managers and customers(students) in the beginning. Users must register first and then they need to go for complete verification.

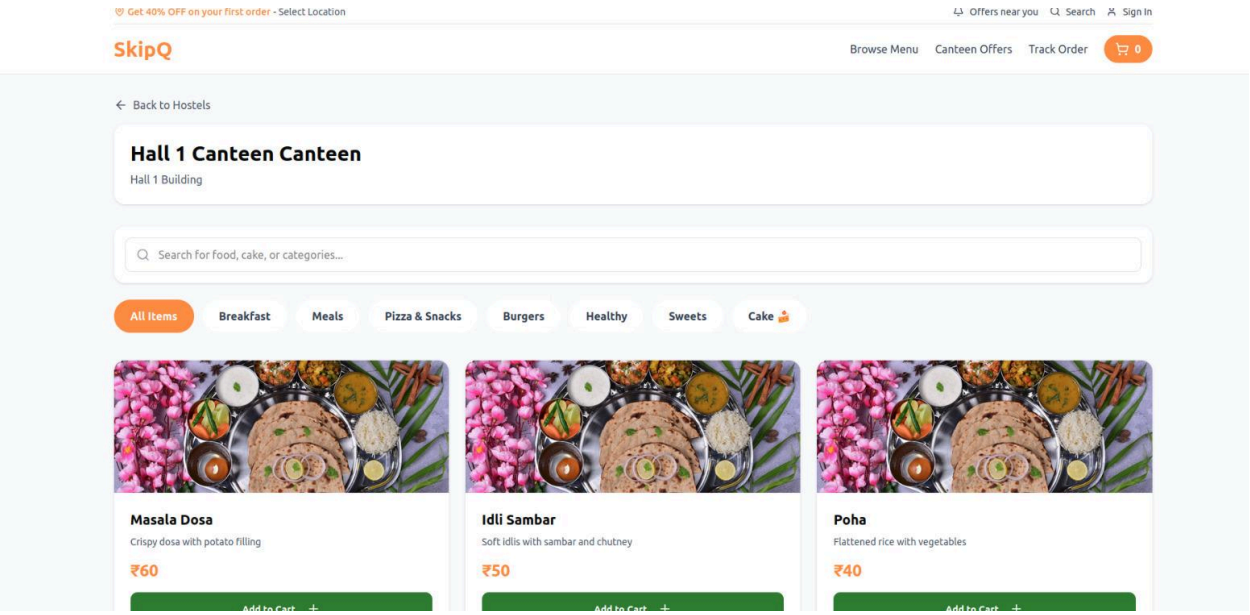


- Customer Interface - It provides various options for the customers like **Profile, Search Bar, Recommendations, Select Canteen, Current orders, History**. In the Profile we see **Edit Profile, Coupons, Wallet** etc.





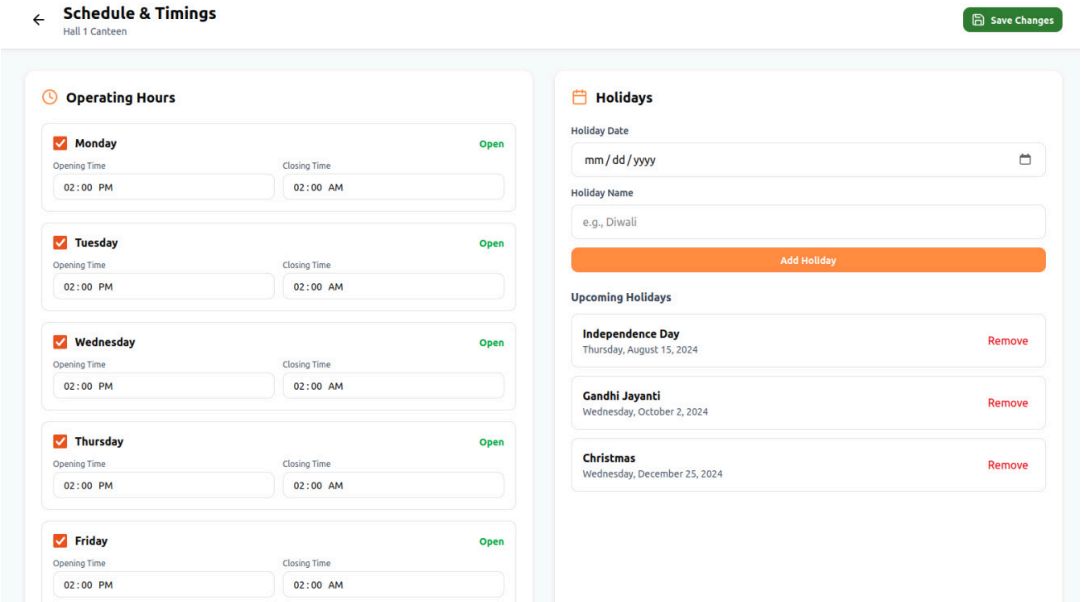
- Canteen Manager/manager Interface - It is the home screen for canteen managers showing

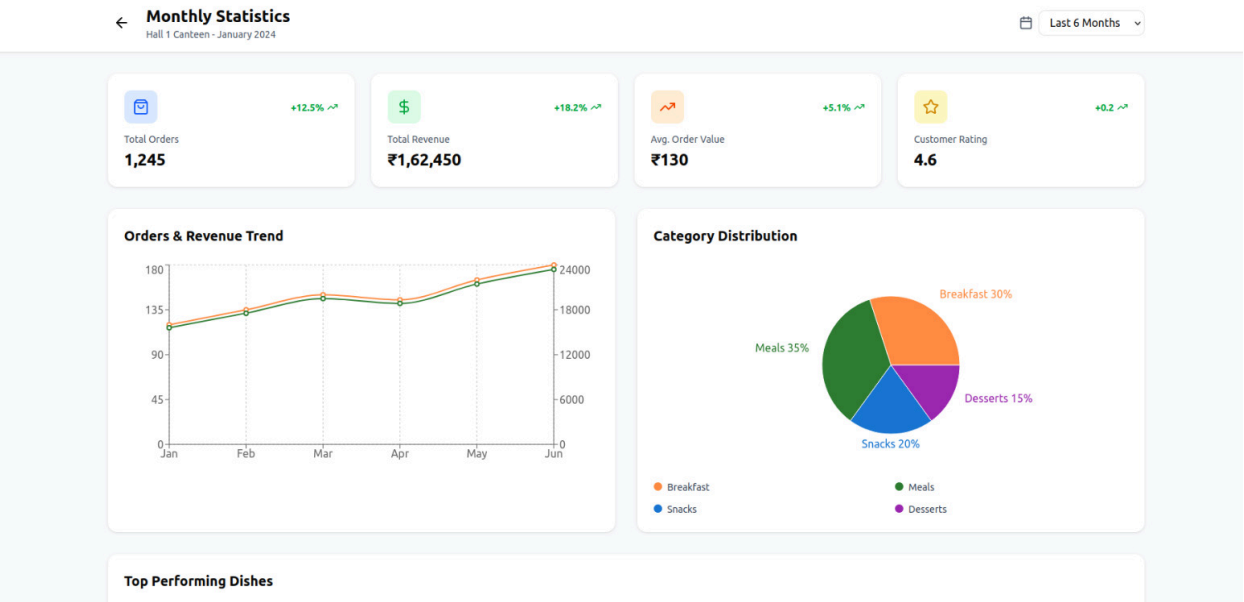


tasks like **Update order time for items, Manage Orders, New order requests, Pending Orders,** and **Update menu**(items and their prices), **Give coupons and discounts.**

- **Admin Dashboard -** A web-based interface (optimized for Desktop) used to onboard new canteens, view global platform analytics, and manage user reports.

(Note - The pictures shown here may not represent the final product.)





3.1.2 Hardware Interface

- **User Devices -** The Webapp will be provided for the user with internet usage on devices like smartphones, tablets and laptops. Thus, it will be easy for simple ordering.

- **Manager Devices -** Webapp would be available the same as above. The canteen managers can use mobile devices like smartphones for their convenience to update Order Queues, Update order time, Accept New orders, and update the menu(items and prices).

- **Admin Devices -** Desktop or laptop computers are required for the Admin Dashboard to ensure full access to analytical tools, global maps, and moderation features.

3.1.3 Software Interfaces

- We shall have three types of interfaces: User, Manager, and Admin.

- The backend infrastructure supporting the Manager and Admin components shall be hosted on a UNIX-based operating system. The frontend shall be accessible on mobile devices and web browsers for managers and web browsers only for admins.

- The user-side components shall be accessible on modern web browsers and their mobile applications, like Chrome, Firefox, Safari and Edge, etc.

- The database management system used will be PostgreSQL or MongoDB:
  – The user database shall store all authentication details and personal information of users.
  – The canteen database shall store the canteen manager's data, including authentication information and other canteen-specific details.
  – The admin database shall store admin credentials and activity logs.
  – The dishes database shall store data like the unique ID of the dish, the unique ID of the manager, price, canteen where it is served, availability, discount, etc.
  – The payment database shall store all the information related to payments.
  – The order database shall store all the orders processed by the system. Orders shall be identified by unique IDs.

3.2 Functional Requirements

3.2.1 Create Account

- Two options should be there for signup - as user (student) or as canteen manager.

- The students must use an IITK email address for registration and managers can use their normal email address.
- Both of them need to enter a password with the email address.

- A 'remember me' option shall be offered to keep the user logged in for 31 days.

- A verification email should be sent to the email address provided for registration.

- Include a 'Forgot Password' option asking for some details in case the user forgets his password.

3.2.2 Customer Functionalities

- **Canteen Options -** The user can choose any canteen to place order from. The page shall show if the canteen is open or closed and also its regular timings.

- **Category Selection -** After choosing a canteen, users can choose food categories from the menu and place order.

- **Order history -** Users can use their order history for easy reordering and to compare earlier prices.

- **Dine-in, takeaway and delivery options:** The user can opt for dine-in at the canteen, can takeaway(packaging) and deliver foods.

- **Estimated time for order:** The system displays preparation time for the selected order.

- **Cake reservation:** Separate option for cake reservation for birthdays and events.

- **Rating and Review System:** Users can rate canteens and food items. They can give reviews on food items and canteen services.

- **Payment options:** Customers can pay using UPI, PhonePe, GPay, Paytm or UserWallet.

- **Favourites:** Customers can add food items from a specific canteen to their favourites list so that they can access them easily for the future.

- **Monthly-stats:** The application also shows the user's monthly expenditure.

- **Wallet:** The user can transfer money from his account to **SkipQ Wallet**.

- **Profile Management:** User can edit profile details including updating photo, username, password, setting/resetting Wallet PIN, or logging out.

3.2.3 Manager functionalities

- **Order Notifications:** New orders are notified to the manager through notifications that can be accepted and rejected. Rejection automatically triggers a refund to the user's wallet.

- **Order-Queue:** Options to accept, reject and prioritise orders to maintain convenience.

- **Menu-management:**
  -Add new dishes
  -Remove old dishes
  -Change prices
  -Update current availability of dishes
  -Add photos under dishes

- **Discount and Coupons:** Options to offer discounts on selected items and offering coupons on conditions that can be redeemed.

- **Clock and Calendar:**
  - Update canteen time and close/open status which is visible to customers.
  - Mark upcoming holidays on which the canteen will be closed.

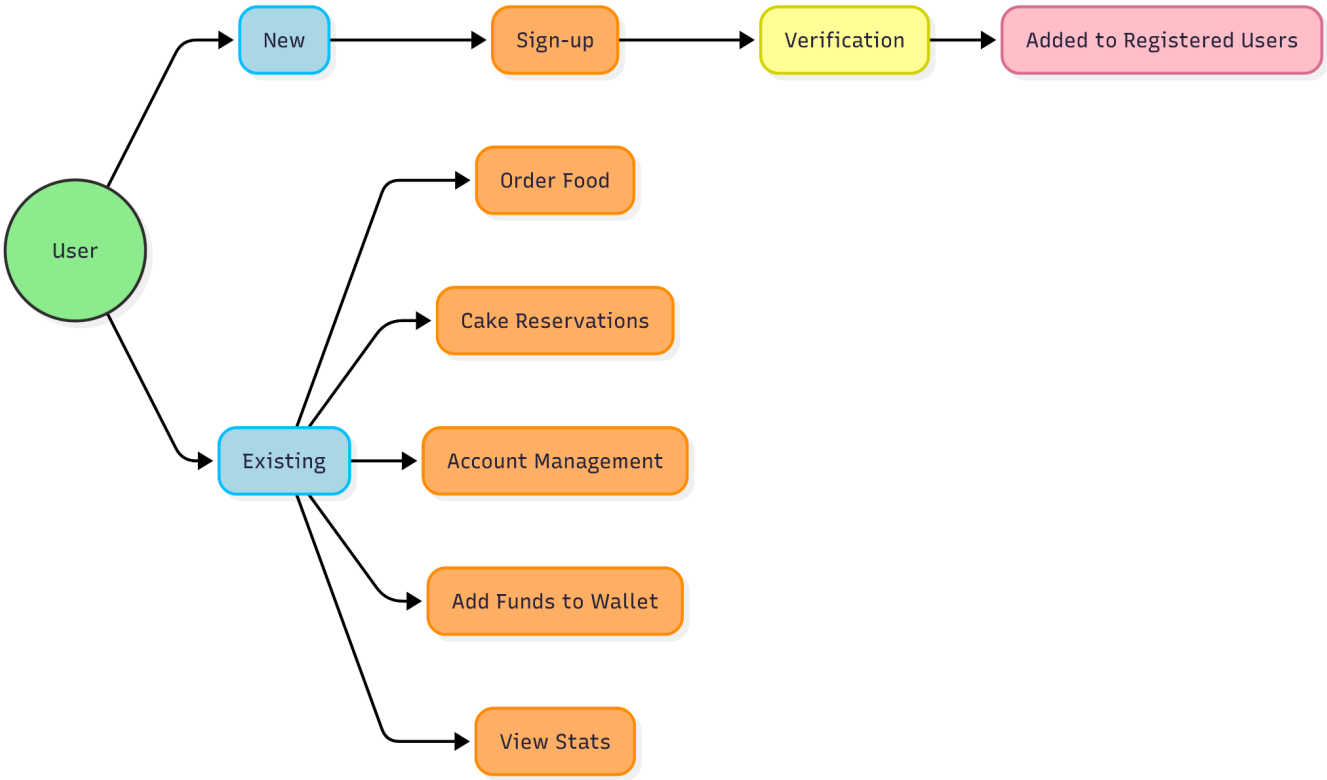
- **Monthly Stats:**
  -Order statistics
  -Earning statistics
  -Most ordered dish of the month

- **Account Management:** Manager can edit profiles like update contact details, change QR code, photos, username or password and logout.

3.2.4 Admin functionalities

- **Canteen Verification:** Capability to approve or reject new canteen registration requests.
- **User Management:** Ability to ban or suspend users/managers who violate terms (e.g., spamming, fake orders).
- **Global Analytics:** View total platform revenue, total active users, and system health status.
- **Content Moderation:** Ability to remove inappropriate reviews or menu items reported by users.
- **Broadcast Notifications:** Send system-wide alerts (e.g., "Server maintenance at 2 AM" or "Platform scheduled for downtime").

3.3 Customer Use Case Model



3.3.1 Use Case #1: Place Food Order and Track Preparation (UC-C01)

Author: Aarya, Usnish

Purpose: To allow the customer to select items from a specific canteen, view real-time wait estimates, and finalize the purchase.

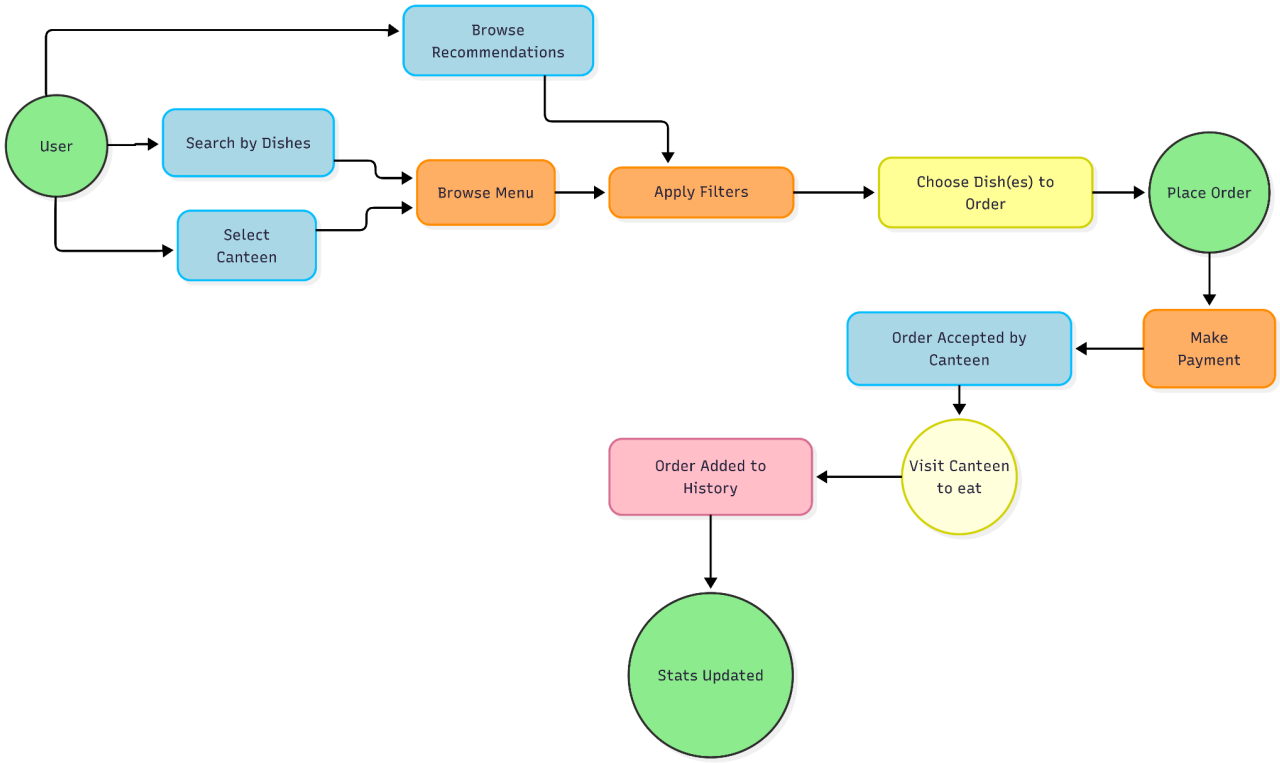
Requirements Traceability: Canteen Options, Category Selection, Estimated time for order, Payment options, Dine-in/takeaway/delivery options.

Priority: High (Core functionality)

Preconditions: Customer must be logged into the SkipQ system with a verified IITK email.

Post conditions: Order is successfully logged in the system; customer balance is updated; Canteen Manager receives a notification.

Actors: Customer (Primary), Payment Gateway (System), Canteen Manager (Secondary).



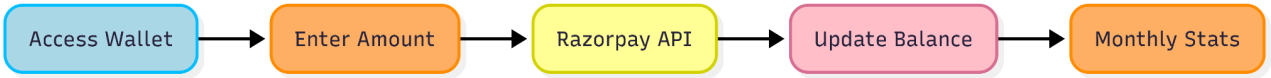
Exceptions:

- Insufficient funds in SkipQ Wallet.
- Canteen closes while the user is browsing the menu.
- Payment gateway timeout/failure.

Includes: UC-C02 (Manage Wallet) if funds are insufficient.

Notes/Issues: Need to ensure the "Wait Time" updates dynamically if a manager changes it while the user is on the checkout screen.

3.3.2 Use Case #2: Manage SkipQ Wallet and Payments (UC-C02)



Author: Harpuneet

Purpose: To enable the customer to add funds, view balance, and track their monthly expenditure statistics.

Requirements Traceability: Payment options, Wallet, Monthly-stats.

Priority: High

Preconditions: User is logged in and has a valid payment method (UPI/NetBanking).

Post conditions: Funds are successfully added to the internal SkipQ Wallet; transaction history is updated.

Actors: Customer, Razorpay API (External System).

Exceptions:
- The bank server is down.
- The user cancels the transaction mid-way.

Includes: None.

Notes/Issues: The system must strictly follow the transition from "Fake Wallet" to "Razorpay API" as per design constraints.

3.3.3 Use Case #3: Submit Review and Manage Favourites (UC-C03)

Author: Pratham

Purpose: To allow customers to provide feedback on their experience and save preferred items for faster reordering.

Requirements Traceability: Rating and Review System, Favourites, Order History.

Priority: Medium

Preconditions: The order must be marked as "Completed" by the canteen.

Post conditions: Review is visible to other users; item is added to the user's "Favourites" list.

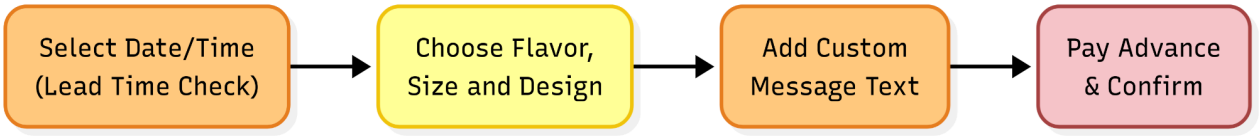
Actors: Customer.

Exception: User tries to review an order that was cancelled or not yet completed.

Includes: None.

Notes/Issues: Reviews must be moderated or have a reporting mechanism to prevent spam.

3.3.4 Use Case #4: Cake Reservation for Events (UC-C04)



Author: Sourabh, Surya

Purpose: To allow customers to pre-order and reserve cakes for specific dates and times, including customization options like messages or flavors, ensuring availability for birthdays or events.

Requirements Traceability: Cake reservation, Payment options, Profile Management (Order History).

Priority: Medium – While not a daily core function like food ordering, it is a high-value specialized service for the campus community.

Preconditions:
- The customer is logged in.
- The selected canteen offers bakery/cake services.
- The reservation is made within the canteen's required lead time (e.g. 6 hours in advance).

Post conditions:
- The reservation is logged in the "Order History."
- The Canteen Manager is notified of a scheduled future task.
- Payment is processed or a deposit is held.

Actors:
- Customer (Primary).
- Canteen Manager (Secondary/Receiver).
- Payment Gateway (System).

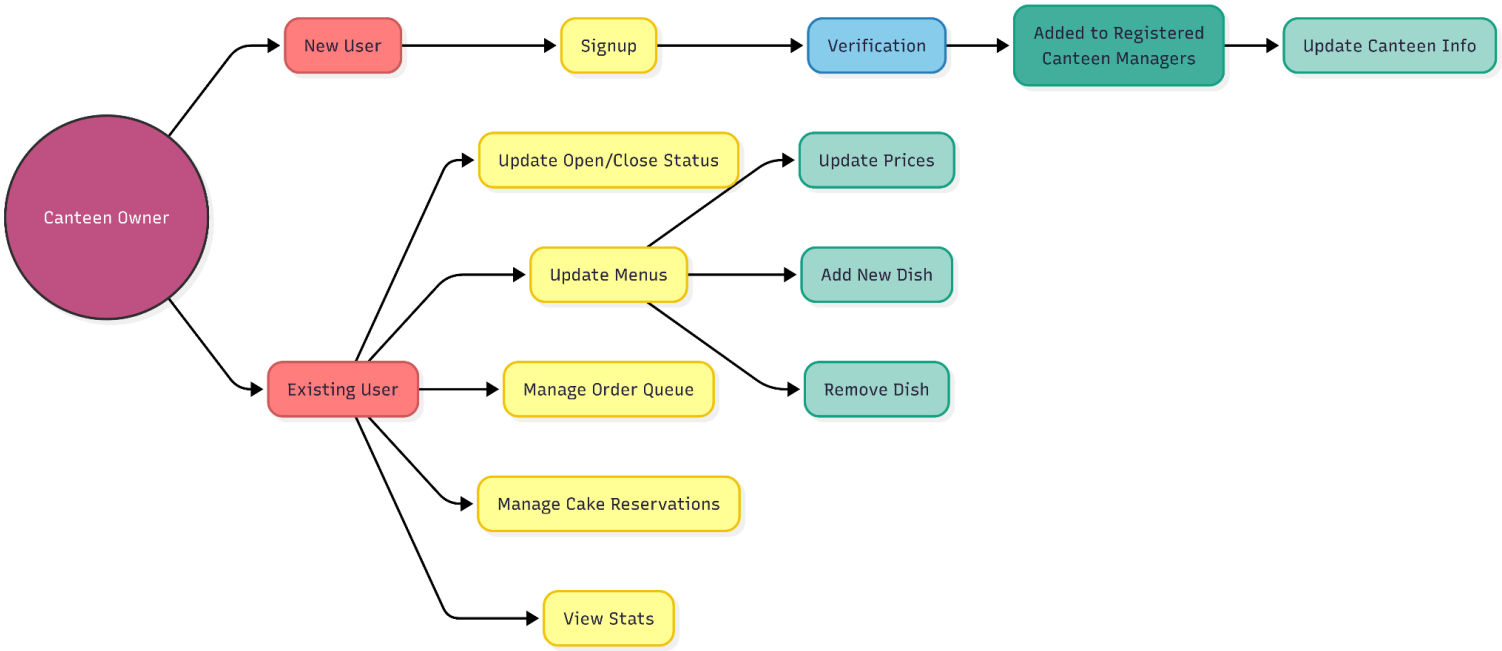
Exceptions:

- **Invalid Lead Time:** Customer tries to book a cake for within 2 hours when the canteen requires 6-hour notice.
- **Customization Limit:** The text for the cake message exceeds character limits.
- **Date Conflict:** The selected reservation date falls on a marked canteen holiday.

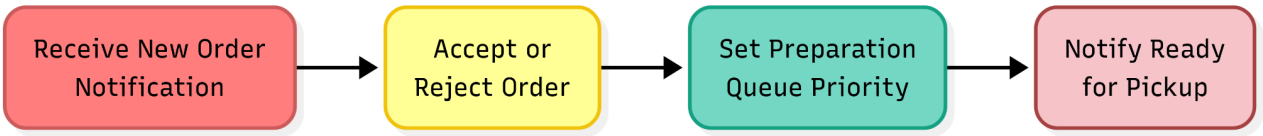
Includes: UC-C02 (Manage Wallet) For payment processing.
Notes/Issues:

- Need to clarify if the user can cancel a cake reservation and if there is a "Cancellation Fee" once the manager accepts it.
- The UI must include a specific "Calendar Picker" that greys out unavailable dates or holidays.

3.4 Canteen Manager Use Case Model



3.4.1 Use Case #1: Live Order Queue Management (UC-M01)



Author: Aayush

Purpose: To receive, prioritize, and update the status of incoming customer orders in real time.

Requirements Traceability: Order Notifications, Order-Queue, Facilitate communication through status changes.

Priority: High

Preconditions: Manager is logged in and the Canteen status is set to "Open."

Post conditions: Customer is notified of status changes (Accepted/Ready); order is moved to "History" upon completion.

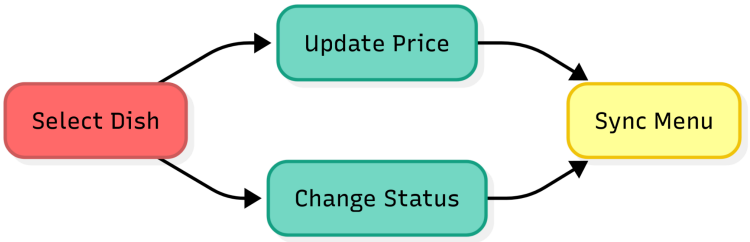
Actors: Canteen Manager, SkipQ Notification Service (System).

Exceptions:
- Manager rejects an order (must provide a reason, e.g., "Out of Stock").
- Network failure prevents real-time status updates via WebSockets.

Includes: None.

Notes/Issues: High volume of orders during peak IITK lunch hours (12 PM - 2 PM) requires the UI to be extremely responsive.

3.4.2 Use Case #2: Dynamic Menu and Pricing Control (UC-M02)



Author: Jashan, Jatin

Purpose: To allow managers to modify the menu, update item availability, and manage discounts/coupons.

Requirements Traceability: Menu-management, Discount and Coupons, Update current availability.

Priority: High

Preconditions: Manager is logged in.

Post conditions: Changes are immediately reflected on the Customer interface.

Actors: Canteen Manager.

Exception: Manager attempts to delete a dish that is currently part of an active (uncompleted) order.

Includes: None.

Notes/Issues: Need a "Bulk Update" feature for prices during festivals or special campus events.

3.4.3 Use Case #3: Operational Scheduling and Analytics (UC-M03)

Author: Ashish

Purpose: To set operating hours, mark holidays, and review monthly earning/order statistics.

Requirements Traceability: Clock and Calendar, Monthly Stats, Account Management.

Priority: Medium

Preconditions: Manager is logged in.

Post conditions: Canteen visibility is toggled based on the schedule; monthly reports are generated.

Actors: Canteen Manager.

Exception: Manager forgets to "Open" the canteen on the app, leading to zero orders.

Includes: None.

Notes/Issues: The "Most ordered dish of the month" stat should be linked to the ML recommendation engine for future features.

3.5 Admin Use Case Model

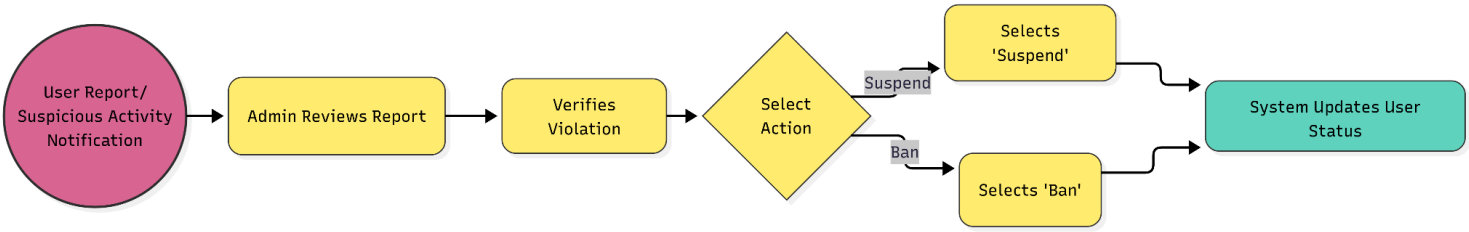
3.5.1 Use Case #1: Onboard New Canteen (UC-A01)

- **Author**: Harpuneet
- **Purpose**: To verify and activate a new canteen in the system.
- **Requirements Traceability**: Canteen Verification.
- **Priority**: High

- **Preconditions**: Canteen Manager has submitted a registration request with documents.
- **Post conditions**: Canteen becomes visible to students in the app.
- **Actors**: Admin.

3.5.2 Use Case #2: Manage User Violations (UC-A02)

- **Author**: Harpuneet
- **Purpose**: To suspend or ban users/managers for policy violations.
- **Requirements Traceability**: User Management, Content Moderation.
- **Priority**: Medium
- **Preconditions**: Admin receives a user report or detects suspicious activity.
- **Post conditions**: User access is restricted or revoked.
- **Actors**: Admin.



4. Other Non-functional Requirements

4.1 Performance Requirements

4.1.1 System Response Time

Requirement: The system shall respond to user actions (such as loading menu items, applying filters, or viewing order status) within **3 seconds** under normal operating conditions and within **5 seconds** during peak hours.

Rationale: Fast response time is crucial to minimize user frustration and allow students to place orders quickly, especially when they are under academic pressure. Delays in interaction can defeat the purpose of reducing queues.

4.1.2 Order Status Refresh Rate

Requirement: The system shall refresh order status (e.g., “Preparing”, “Ready”, “Completed”) at least once every **15 seconds** without requiring manual page reload.

Rationale: Frequent status updates would display the accurate status of the order and help students time their visit to the canteen accurately.

4.1.3 System Availability During Operating Hours

Requirement: The system shall maintain at least **99% availability** during canteen operating hours.

Rationale: Frequent downtime or crashing would force students back into physical queues, defeating the core objective of **SkipQ**.

4.1.4 Concurrent Users Handling

Requirement: The system shall be able to support at least **500 concurrent users** (students and canteen managers combined) during peak hours without significant degradation in performance.

Rationale: During peak hours such as lunch breaks or exam periods, many students may access the system simultaneously. Supporting high concurrency ensures stable performance without crashes or slowdowns.

4.2 Safety and Security Requirements

4.2.1 Wallet Transaction Authorization

Requirement: All payments done through the wallet shall require a **unique wallet PIN** to authorize the transaction.

Rationale: This feature adds a critical security layer to protect user funds if the physical

device is compromised, even with an active user session. It prevents an unauthorized party who gains physical access from doing transactions.

4.2.2 Payment Validation

Requirement: The order should be placed only after the transaction has been verified, i.e, money has been deducted from the user's account and has been credited to the canteen manager's account.

Rationale: This ensures that order is not processed in case of unpaid or failed transactions. This also mitigates food wastage and monetary loss for canteen managers.

4.2.3 Data Backup

Requirement: The system shall regularly take backup of data.

Rationale: In case of system failure, critical data such as wallet information, transaction history must not be lost.

4.2.4 User Authentication

Requirement: User authentication shall be conducted via email identification, utilizing a One-Time Password (OTP) mechanism. Furthermore, all passwords should be encrypted using secure cryptographic algorithms.

Rationale: This ensures managership of given email address and prevents fake account creation. This also prevents unauthorized access of the user's account.

4.2.5 Data Privacy

Requirement: All personal information such as wallet information, transaction history, etc. should be securely saved such that it is not exposed to unauthorized access.

Rationale: This ensures that user's sensitive information remains confidential and cannot be accessed by others.

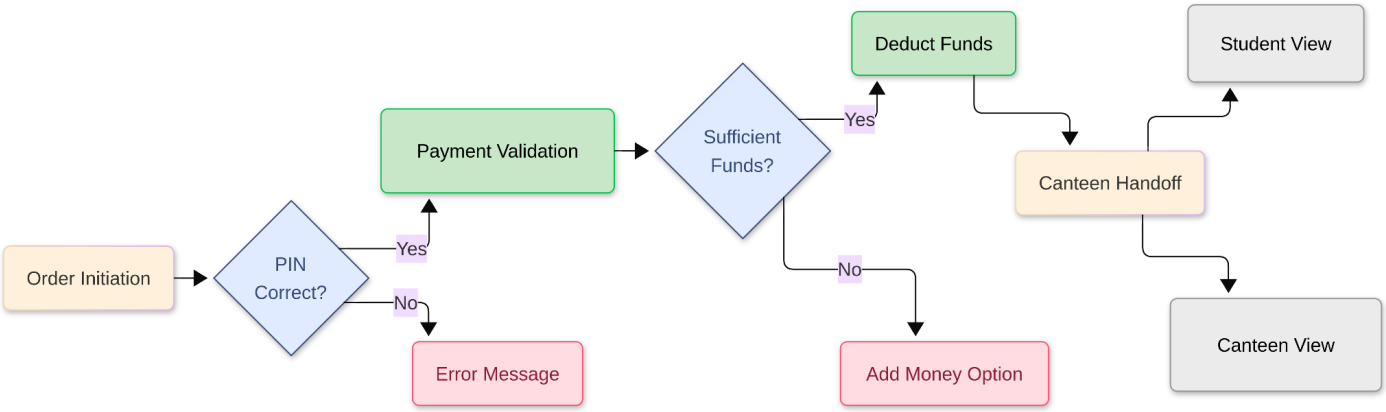


Figure: Flowchart of Payment System

4.3 Software Quality Attributes

4.3.1 Ease of Use

Requirement: New users (students) shall be able to place an order using the SkipQ system within 2 minutes without external assistance.
Ease of use is prioritized over ease of learning, as most users will interact with the system frequently and under time constraints.

How it shall be achieved:

- The user interface should be simple and intuitive, inspired by familiar food-ordering platforms.
- Clear navigation, readable fonts, and logical grouping of menu items shall be used.
- Filters and sorting options shall help in decision-making.

***Verification*:** Usability testing shall be conducted with sample users to measure task completion time.

4.3.2 Maintainability

***Requirement*:** The SkipQ system shall allow developers to identify, modify, and fix software defects within **one working day** without compromising existing functionality.

How it shall be achieved:

- The system shall have separated frontend, backend and database layers.
- Clear coding standards and meaningful variable naming should be followed.
- Robust error reporting mechanisms should be implemented to simplify debugging and issue tracking.

Verification: Maintenance effort can be evaluated by measuring the time required to locate fix known defects during testing and post-deployment phases.

4.3.3 Adaptability

Requirement: The SkipQ system shall support the integration of new features (such as additional payment methods, new food categories, or recommendation system) within **two working days**, without requiring major changes to existing system components.

How it shall be achieved:

- Core system functionality shall be separated from optional and extendable features to prevent changes in one module from affecting others.
- Feature-specific logic (e.g., payment processing, recommendations) should be implemented as **independent modules**.

5. Other Requirements

5.1 Legal Requirements and Copyrights

All user data collected by the system shall be stored in data centers located within the country to comply with national data protection and privacy regulations. The copyright and management of the source code and all project documentation shall remain with the system developers.

5.2 Data Recovery Requirements

5.2.1 Disaster Recovery

A complete, restorable backup of the entire database must be performed daily.

5.2.2 Data Migration

The database schema and data must be designed to allow for a seamless transition between a test/staging environment and the production environment.

5.3 Authentication

Permission will be required to send OTPs for authentication via automated emails or SMS. This shall ensure secure user verification during the signup and login processes.

Appendix A – Data Dictionary

A.1 User Database

Attribute	Description	Requirements
ID	Unique ID of the user	Unique for each user
Personal Details	Personal information of the user	String
Password	Password of the user	Encrypted string
Wallet	User's wallet balance	Numeric value
Wallet PIN	Security code for transactions	Encrypted Hash (4-6 digits)
Email	Email of the user	IITK Email
Fav Dish	User's favorite dish	String
Current Orders	List of ongoing orders	List of orders

A.2 Canteen Database

Attribute	Description	Requirements
ID	Unique ID of the canteen	Unique for each canteen
Name	Name of the canteen	String
Canteen ID	ID of the canteen	Unique for each canteen
Password	Password to access the canteen database	Encrypted string
Menu	List of items available in the canteen	List of dishes
Order	Current orders placed	List of dishes
Queue	Current order queue	List of dishes
Previous Order Details	Details of past orders	List of orders
Opening Time	Daily opening time of the canteen	Time format (HH:MM)
Closing Time	Daily closing time of the canteen	Time format (HH:MM)
Contact Details	Contact information of the canteen	String

A.3 Dishes Database

Attribute	Description	Requirements
ID	Unique ID of the dish	Unique for each dish
Price	Price of the dish	Numeric value
Canteen Name	Name of the canteen offering the dish	String
Availability	Availability status of the dish	Boolean
Discount	Discount offered on the dish	Numeric value (Percentage)
Photo	Image of the dish	File/Image
Other Miscellaneous Details	Additional information about the dish	String

A.4 Payments Database

Attribute	Description	Requirements
ID	Unique ID of the payment	Unique for each payment
User ID	ID of the user who made the payment	Must be a valid ID
Order ID	ID of the order for which the payment was made	Must be a valid ID
Amount	Amount of the payment	Numeric value
Time	Time at which the payment was made	Time
Payment Method	Method used for the payment	String
Status	Whether the payment was successful or not	Boolean

A.5 Order Database

Attribute	Description	Requirements
ID	Unique ID of the order	Unique for each order
User ID	ID of the user who placed the order	Must be a valid ID
Current Status	Current state of the order	Enumerated Value
Booking Time	Time at which the order was booked	Time
Delivery Time	Time at which the order was delivered	Time
Dish and Quantity	List of dishes and their respective quantities	List of pairs
Order Amount	Total amount for the order	Numeric value

A.6 Admin Database

Attribute	Description	Requirements
ID	Unique ID of the Admin	Unique for each Admin
Email	Admin email address	Valid email format
Password	Admin login credentials	Encrypted string
Role level	Permissions for each admin	Enumerated Value
Activity Log	Record of actions taken	List of actions with timestamps

Appendix B - Group Log

Meeting notes and activities will be maintained here

S.No.	Date	Timings	Venue	Agenda
1	06/01/ 2026	14:00 to 16:00	RM Building	Discussed and proposed various project ideas, including: Canteen and Mess Management System, Hostel Management Application.
2	09/01/2026	10:00 to 12:00	RM Building	Finalized the project idea and brainstormed specific requirements and functionalities.
3	10/01/2026	18:00 to 19:00	L-7	Met with the professor to present the project idea and seek suggestions on potential technologies to use.
4	14/01/2026	22:00 to 23:00	RM Building	Reviewed and discussed the SRS (Software Requirements Specification) template.
5	20/01/2026	12:00 to 13:00	L-7	Explored different technologies to implement in the project and planned a meeting with the mentor for additional guidance.
6	23/01/2026	16:00 to 17:00	Google Meet	Discussed changes suggested by mentor
7	24/01/2026	19:00 to 20:00	Google Meet	Finalised the Document